

■ BloodBridge — Comprehensive Technical & System Documentation

Enterprise Blood Donation & Emergency Allocation Platform | Tamil Nadu Scope

1. Executive Overview

BloodBridge is an enterprise-grade healthcare web application designed to connect voluntary blood donors with patients in emergency need through verified blood camps and hospital destinations. Built following Feature-Sliced Design principles, BloodBridge strictly enforces regional Tamil Nadu district isolation, clinical ABO/Rh compatibility rules, urgency-based priority allocation, and real-time inventory synchronization.

Category	Technologies Used
Frontend Framework	React 19, TypeScript 6, Vite 8
Styling & Theme Engine	Tailwind CSS v4, Lucide Icons, Custom CSS Color Engine (3 Themes)
Backend & Database	Firebase Realtime Database v12, Firebase Auth (Google OAuth & Email/Password)
Routing & State	React Router v7, React Context (Auth, Toast, Theme, RBAC)

2. Core Features & Capabilities

A. Tamil Nadu District Scope & Geofenced Broadcasting

- Scoped across all **38 Tamil Nadu Districts** (Chennai, Madurai, Coimbatore, Salem, Thanjavur, etc.).
- Emergency blood requests raised for a patient at a destination hospital are broadcast **strictly to voluntary donors and blood camps in that destination hospital's district**.

B. Clinical ABO/Rh Blood Compatibility Matching Engine

- Implements strict clinical ABO and Rh factor compatibility matching rules (e.g. O- universal donor, B+ can donate to B+ and AB+, Rh-null, Bombay blood group support).
- Incompatible blood groups are automatically blocked in UI and guarded by backend database checks.

C. Urgency-Based Priority Ordering (Critical -> Urgent -> Normal)

- Emergency blood requests are prioritized across all dashboards: ■ **Critical (Immediate)** first, ■ **Urgent (Within 12h)** second, and **Routine Normal** third.

D. 7-Day Expired Requests Archive & Admin Database Cleanup

- Active requests stay on live donor dashboards for 7 days. Requests older than 7 days automatically move to an **Expired Requests Archive** section.
- Includes filter options ('All Expired', 'Still Not Closed', 'Closed') and a quick 1-click 'Close Case' modal for admins.

E. Voluntary Individual Donor FCFS System & Recovery Lock

- Donors can commit maximum 1 unit per request on a First-Come, First-Served basis.
- Completing a donation places the donor on a mandatory **56-day WHO recovery lock**.
- Requesters get live donor phone lookup & clickable tel links; Admins have 1-click Rebroadcast tools.

F. Google OAuth Sign-In & Mandatory Profile Onboarding

- 1-Click 'Continue with Google' authentication.

- Automatically prompts new Google users with a **Complete Donor Profile Onboarding Modal** to select their Blood Group, District, and Contact Phone.

G. Universal Profile & Location Settings with Account Deletion

- All roles can change their Tamil Nadu district anytime when moving.
- Danger Zone account deletion requires typing exact confirmation string: *'I am deleting my account'*.

H. Multi-Theme Engine & Blood Core Switcher

- Supports 3 100% solid high-contrast themes: **Crimson Dark**, **Executive Light**, and **Emerald Health**.

3. 5-Stage Transactional Workflow Engine

Stage	Description & Observability Guardrails
1. Registered	Request created by user/hospital with atomic reference BB-YYYY-NNNNN.
2. Verified	Verified by Blood Camp / Hospital Manager; broadcast to district donors.
3. Matched	Compatible voluntary donors or camp stock matched to request.
4. Donated	Blood units delivered; auto-deducts inventory & sets 56-day donor recovery lock.
5. Closed	Case completed with mandatory closure notes. Immutable state.